

<https://www.youtube.com/watch?v=1dPHY4u7wRM>
https://www.youtube.com/watch?v=k1w4u_ZOTXg

Topic	URL
TMap important script	URL - Articles
TMap Best practices	https://www.servicenow.com/community/developer-blog/transform-map-best-practices/ba-p/2276233
scripts on fieldMap and Transform script	https://docs.perspectium.com/display/Krypton/Update+ServiceNow+transform+map+field+scripts+for+Freshservice
SNOW_TMap docs	https://docs.servicenow.com/bundle/washingtondc-integrate-applications/page/script/server-scripting/concept/CreatingNewTransformMaps.html
SNOW_Data source docs	https://docs.servicenow.com/bundle/washingtondc-integrate-applications/page/administer/import-sets/concept/_DataSources.html
Concerns on NULL Values	https://www.servicenow.com/community/developer-forum/confusion-over-null/m-p/1932483
If condition based on NULL field	https://www.servicenow.com/community/developer-forum/if-condition-based-to-a-null-field/m-p/2002229

Excel : auto expand the columns - alt + H, O, I
concatenate the column - =conactenate("some text",column name)

What is data sources and how it will work?

A data source contains data of the resources like external data sources such as excel, CSV, XML, JDBC and HTTP/FTP, that can be stored in the import set table with the help of transform map it can be transform in to the target table.

import set table is nothing but the source table.

create a data source :

Data Source Demo source

* Name: Demo source

Import set table label: Demo source

* Import set table name: u_demo_source

* Type: File

Format: Excel (.xlsx/.xls)

Zipped: ☐

Sheet number: 1

Header row: 1

Use Batch Import: ☐

Application: Global

File retrieval method: Attachment

Update

select the type as you required, here using as file

format as excel

sheet number is based on the excel sheet

header row is which row contains the header see the below image

	A	B	C	D	E
1	Number	Caller	Category	Short description	Description
2	TEST0001	Abel Tuter	Inquiry / Help	This is test for short description on - Abel Tuter	This is test for description on the user is - Abel Tuter , category is - Inquiry / Help
3	TEST0002	Abraham Lincoln	Inquiry / Help	This is test for short description on - Abraham Lincoln	This is test for description on the user is - Abraham Lincoln , category is - Inquiry / Help

here the header row is 1

Note, there is lot of different type and format(try to explore it)

file retrival method is attachment because need to attach the excel in the data source,
file retrival method has different categories like

Application Global ⓘ

File retrieval method Attachment ▼

- Attachment
- FTP
- FTPS (Auth SSL) [not recommended]
- FTPS (Auth TLS) [not recommended]
- FTPS (Implicit SSL) [not recommended]
- FTPS (Implicit TLS) [not recommended]**
- HTTP
- HTTPS
- SCP
- SFTP

once the excel file is attached in the data source then click "load 20 records" then it will upload all 20 data in the import set table "u_demo_source" table

Progress

Name	ImportProcessor
State	Complete
Completion code	Success
Message	Processed: 5, inserts 5, updates 0, errors 0, empty and ignored 0, ignored errors 0 (0:00:01.127)

Next steps...

[Import sets](#) Go to the import sets for this data load

[Loaded data](#) Go to the newly imported data inside the staging table: u_demo_source

[Create transform map](#) Create a transform map for the newly staged data

[Import log](#) View the import log

by clicking the loaded data you can see all the records and state as "Test"

Demo sources

Set

Search

All

	Row	Set ▼	State	Target table
	1	ISSET0010001	Test	
	3	ISSET0010001	Test	
	0	ISSET0010001	Test	
	4	ISSET0010001	Test	
	2	ISSET0010001	Test	

Related Links

- [Import Sets](#)
- [Transform Maps](#)
- [Transform History](#)
- [Edit Web Service](#)

it will create all the fields as mentioned in the excel columns

Demo source

1

Number

TEST0002

Short description

This is test for short description on - Abraham Lin

Category

Inquiry / Help

Description

This is test for description on the user is - Abraham Lincoln , category is - Inquiry / Help

Caller

Abraham Lincoln

Update

Delete

Note : it is staging table, something like temporary table once the data is transformed from stage table to target table these staging table data will gets deleted

now those loaded data need to be created in the target table, in this case we can use incident table

Progress

Name

ImportProcessor

State

Complete

Completion code

Success

Message

Processed: 5, inserts 5, updates 0, errors 0, empty and ignored 0, ignored errors 0 (0:00:01.127)

Next steps...

- [Import sets](#) Go to the import sets for this data load
- [Loaded data](#) Go to the newly imported data inside the staging table: u_demo_source
- [Create transform map](#) Create a transform map for the newly staged data
- [Import log](#) View the import log

click "create transform" to create a transform map

Table Transform Map
New record

* Name: Create Incident through Data source_V1

* Source table: Demo source [u_demo_source]

Active: ☒

Run business rules: ☐

Enforce mandatory fields: No

Copy empty fields: ☐

Create new record on empty coalesce fields: ☐

Application: Global

Created:

* Target table: Incident [incident]

Order: 100

Run script: ☐

here "enforce mandatory fields" means even any mandatory fields requires to create a record in the target record, still it will create a record if it is set as "No"

copy empty fields - id field/row is empty in the excel still it will create a record by checking the field

create new record on empty coalesce field - ???

once TMap is created now create the field map,
3 ways : Auto map matching fields, Mapping assist, manually by clicking "New"

Related Links

[Auto Map Matching Fields](#)

[Mapping Assist](#)

[Validate Coalesce Fields](#)

[Transform](#)

[Index Coalesce Fields](#)

[Run Point Scan](#)

Field Maps (5) Transform Scripts Empty reference fields (1)

Source field	Target field
u_category	category
u_number	number
u_caller	caller_id
u_short_description	short_description
u_description	description

if appropriate fields are did not mapped do it on manually.

if we set coalesce field as true, then check for the particular id(unique id) if it is there then it will update or else it will create

script variables :

Variable Name	Types
Source	Glide record object
Target	Glide record object
Map	Glide record object
Log	function
Action	function
Ignore	Boolean
Error	Boolean
error_message	String(output message)
status_message	String(output message)

Transform map script:

3 type of script:

1. Explicit tMap script - in the tMap

e mandatory fields Run script ☒

Copy empty fields ☒

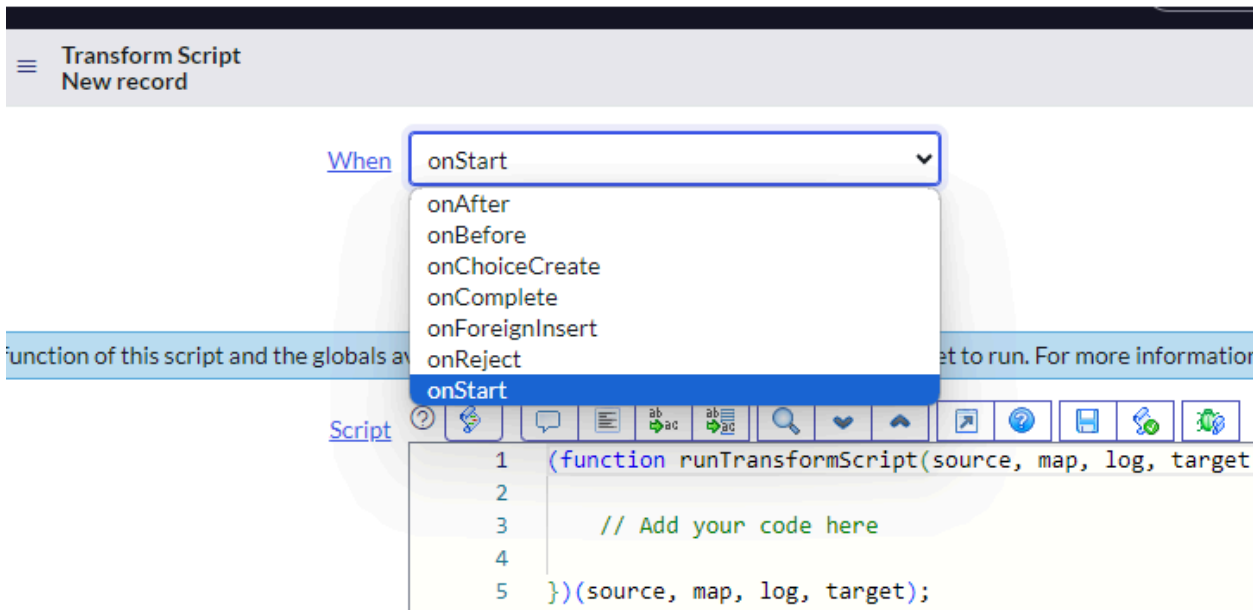
on empty coalesce fields ☒

Script

```
1 (function transformRow(source, target, map, log, isUpdate) {  
2  
3     // Add your code here  
4  
5 })(source, target, map, log, action=="update");
```

2. field map script - inside the field map click "use source script"

3. transform script - it has multiple option



Script Types:

1. Explicit
2. Field Map Script
3. Transformation event scripts
 - **onStart:** The onStart event script is processed at the start of an import run, before any data rows are read.
 - **onComplete:** The onComplete event script is processed at the end of an import run, after all data rows are read and transformed.
 - **onBefore :** The onBefore event script is processed at the start of a row transformation, before the source row is transformed into the target row.
 - **onAfter:** The onAfter event script is processed at the end of a row transformation, after the source row has been transformed into the target row and saved.
 - **onForeignInsert:** The onForeignInsert event script is processed at the start of the creation of a related, referenced record, before the record is created.
 - **onChoiceCreate:** The onChoiceCreate event script is processed at the start of a choice value creation, before the new choice value is created.
 - **onReject:** The onReject event script is processed during the occurrence of a foreign record or choice creation, and the foreign record or choice is rejected ,the entire transformation row is not saved.

onBefore and onAfter runs on each record/ or in each data.

onforiegn insert - in target table the reference field is referred to different table like target table is incident but caller field is referenced to "user" table, if the value is new which it is not present in the user table then it will become like an foreign insert

flow diagram to understand the transform scripts:

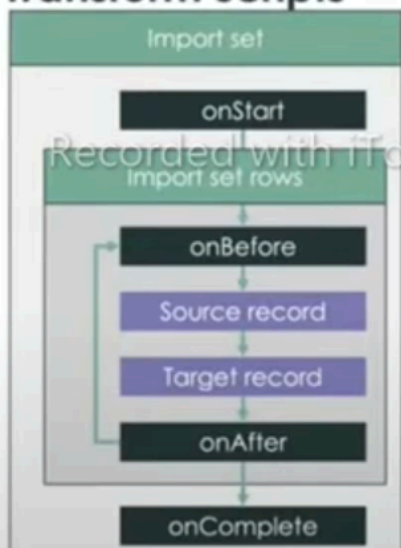
Transform scripts



onStart = Run once at the start of the transform process

Recorded with iTop Screen Recorder

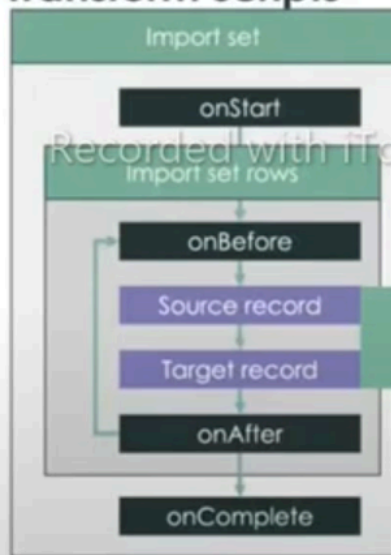
Transform scripts



onAfter = Run AFTER the source row has been transformed and saved

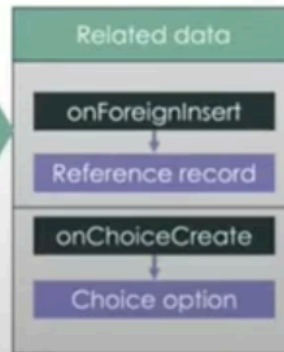
Recorded with iTop Screen Recorder

Transform scripts



onForeignInsert = Run before a referenced record is created

Recorded with iTop Screen Recorder

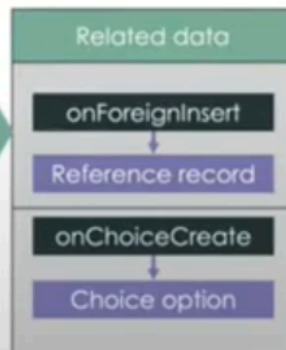


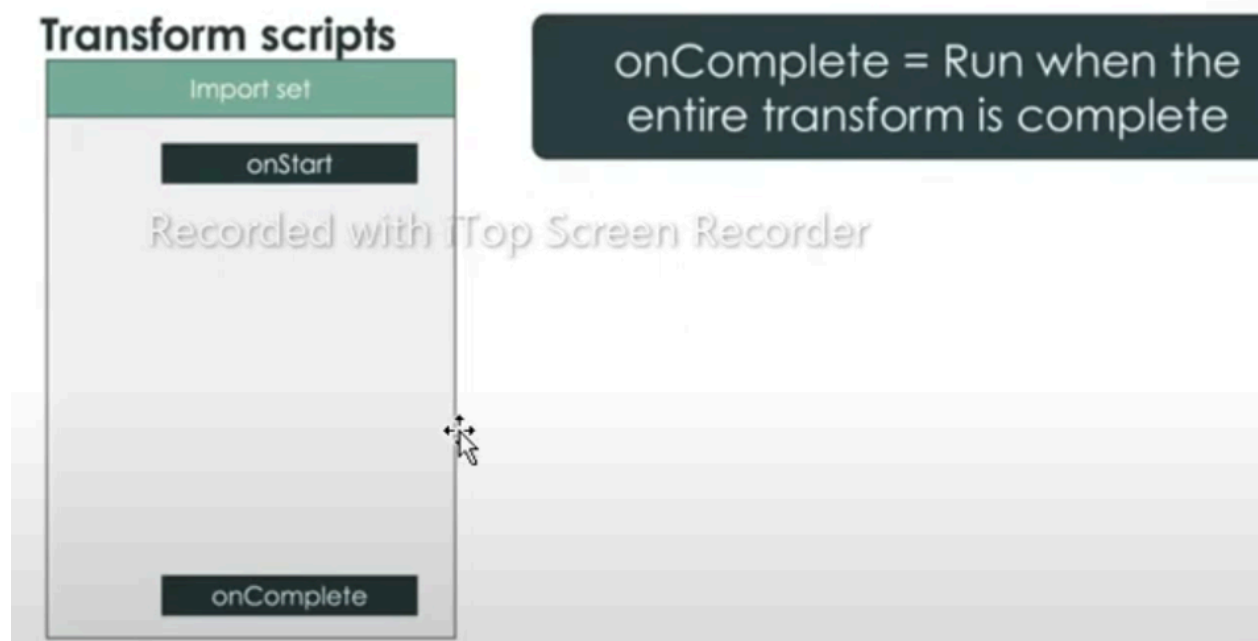
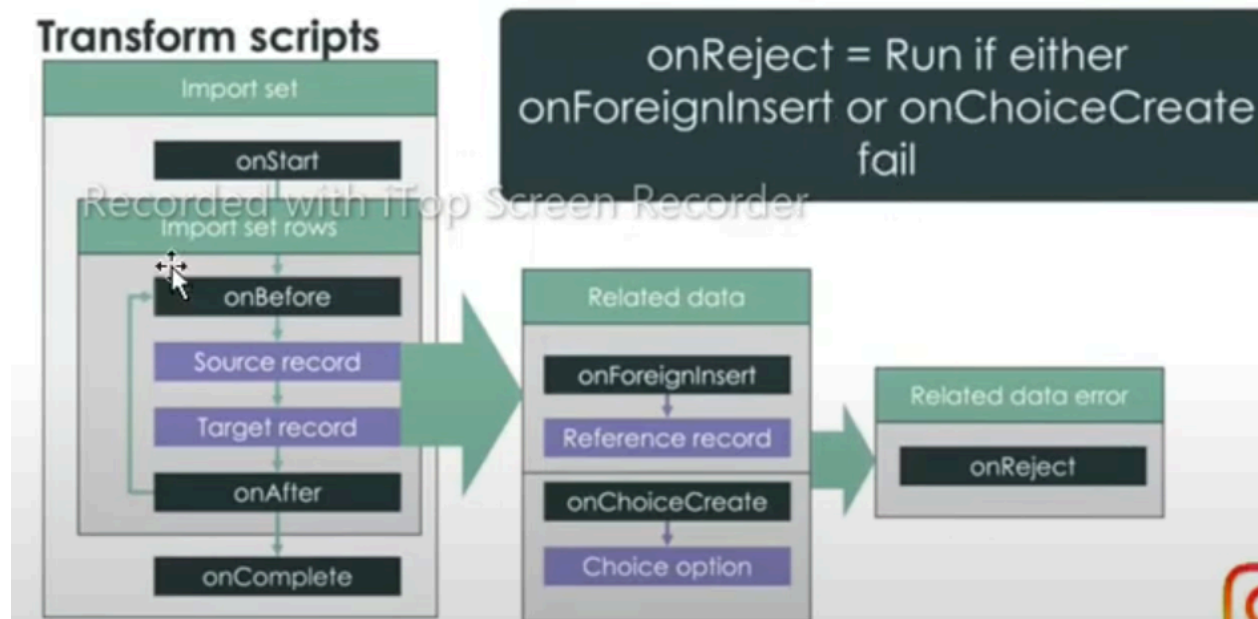
Transform scripts



onChoiceCreate = Run before a new choice list entry is created

Recorded with iTop Screen Recorder





How explicit script works?

use case : if category is hardware automatically change the subcategory as "monitor"

this script will need to write on the tMap by check the checkbox called "Run script"

run business rules ☐

force mandatory fields

No

Copy empty fields ☒

ord on empty coalesce fields ☒

Order

Run script ☒

Script

1

2

3

4

5

6

7

8

9

10

11

12

```

(function transformRow(source, target, map, log, isUpdate) {
    // Add your code here

    if (source.u_category == 'Hardware') {
        target.subcategory = 'monitor';
        /*
        this will update the value in the target record once the source record details is imported
        */
    }
}
)(source, target, map, log, action === "update");

```

field map script:

the field map script will return what kind of values needs to be stored in the source table
all the field map script should be pass in after the "return" keyword

onForiegnInsert():

in the field map, for the choice field and reference field, there will be an option called "choice action"

based on the field which it is mapped into the target table

in choice action it has 3 options :

create - if no records are there it will create the record in reference table and create a record in target table

ignore - if no records are there it will make it as empty and create a record in target table with empty value

reject - if no records are there it will completely reject the row

transform map script:

use case: when the caller is created, need to update the created user record in the user table with the eventQueue()

step 1: call a eventQueue in the transform script and pass it appropriate parameters

When

Application

Active ☒

Order

globals available to it vary depending on when the script is set to run. For more information, see [the Wiki](#). See also the article about the [recommended for](#)

Script

```

1 (function runTransformScript(source, map, log, target /*undefined onStart*/ ) {
2
3     // Add your code here
4     log.info('Foreign insert is happened');
5     gs.eventQueue('User Update', source, source.u_caller);
6
7 })(source, map, log, target);

```

step 2: create a event called "User Update" in the "sysevent_register" table.

Event Registration
User Update

al actions

* Event name

Table

Priority

Fired by

Description

step 3: create a event script in the 'sysevent_script_action' to validate and update the reference record of email id and title

Name	User Update
Event name	User Update
Execution Order	100
Condition script	
Script	? <pre>1 var gr = new GlideRecord('sys_user'); 2 gr.addQuery('name', event.parm1.toString()); 3 gr.query(); 4 if (gr.next()) { 5 gr.email = gr.name + "@gmail.com"; 6 gr.title = "Manager"; 7 gr.update(); 8 }</pre>

step 4: load data and run the transform and see the records which it is inserted in the target table

****** if field map has script : then call the source table field backend value ******

